MICROECONOMICS

MICRO-04

Time: 6 minutes

Outcome: Calculate midpoint price elasticity of demand and interpret the result

Difficulty: Intermediate

Price Elasticity of Demand

THE CORE IDEA

Price elasticity of demand measures how strongly quantity demanded responds to a price change: PED = absolute percentage change in quantity demanded divided by percentage change in price. The midpoint method uses the average of the two quantities and the average of the two prices, so reversing the direction gives the same result. Demand is elastic when PED is greater than 1, unit elastic at 1, and inelastic below 1. Elasticity uses percentages, not raw unit changes or visual slope. Close substitutes, a longer adjustment period, a larger budget share, and a narrower market definition generally make demand more elastic.

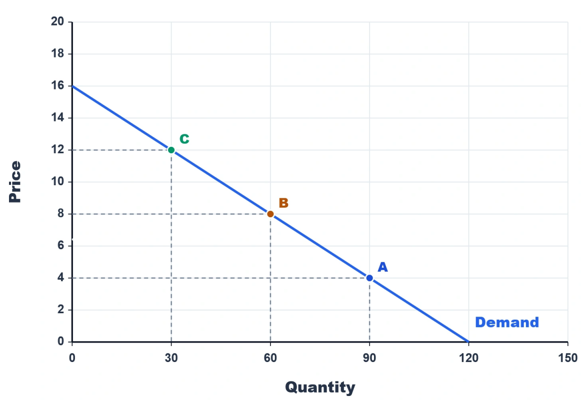
HOW TO RECOGNIZE IT

- Two price-quantity points are provided for a midpoint calculation.
- The problem asks whether the quantity response is elastic, unit elastic, or inelastic.
- Substitutes, time to adjust, budget share, or market scope are changing.
- A linear demand curve has constant slope but different elasticity along the line.
- A horizontal or vertical demand curve represents a limiting elasticity case.

WATCH OUT

A steeper-looking curve is not automatically less elastic. Slope uses the original price and quantity units, while elasticity compares percentage changes and depends on the price-quantity location. Put percentage quantity change in the numerator, use midpoint averages in both denominators, and classify demand using the absolute value of PED.

WORKED EXAMPLE: MIDPOINT ELASTICITY



Between C and B, price falls from \$12 to \$8 while quantity demanded rises from 30 to 60. Using midpoint values, quantity changes by 66.7% and price changes by 40.0%, so PED = 66.7% / 40.0% = 1.67. Because 1.67 is greater than 1, demand is elastic over this interval: quantity responds proportionally more than price changes.

CHECK YOURSELF

Using points B (P = \$8, Q = 60) and A (P = \$4, Q = 90), what is midpoint PED, and how should demand be classified over that interval?

READY? Return to the game and master this concept.